TAPE 16

Documentation

Version 0.9.330

EMR Music Group

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Contents

TAPE 16 is intentionally tape-first: fast decisions, committed recordings, and a clear 16-track surface. This documentation focuses on practical operation inside the app.

Version: 0.9.330

TAPE 16 uses destructive recording by design. When you record onto an armed track, the new recording replaces audio on that track over the recorded range.

1. First Launch And Activation

On first launch, TAPE 16 may show an activation or trial screen.

- Enter your serial number in the format `T16-XXXXXX-XXXXXX-XXXXXX`.
- Use `Activate` for online activation.
- Keep your serial number somewhere safe for future machines or reinstalls.

Your license and entitlement files are stored locally in your user application data folder under `TapeDAW`.

2. Recommended Display Setup

TAPE 16 is a dense hardware-style interface. For the full control surface to remain comfortable, use at least `1920 x 1080` resolution and open the app fullscreen when possible.

Use `Preferences > Global UI Scale` to resize the complete TAPE 16 interface. `Auto` chooses a scale from the display's usable height; manual settings range from `75%` to `150%`. Smaller percentages create more working room on compact laptop displays, while larger percentages improve control size on high-resolution monitors.

The separate transport, text, VU, reel, and side-menu size controls remain available for fine adjustment. If controls are still clipped or hard to reach:

- switch TAPE 16 to fullscreen,
- reduce `Global UI Scale`,
- or use the app on a larger external display.

3. Where Tapes Are Stored

By default, TAPE 16 stores user sessions in:

- macOS primary location: `~/Music/TAPE/TAPE 16/TAPES`
- Legacy or fallback location: `~/Documents/TAPE/TAPE 16/TAPES`

Related folders:

- Tape sessions: `TAPE 16/TAPES`
- Templates: `TAPE 16/TAPE 16 Templates`
- Bounces: `TAPE 16/TAPE 16 Bounces`

You can choose a custom tapes root in the tape library, which is useful for external drives or dedicated session storage.

4. Main Screen Overview

The main screen is laid out like a 16-track tape machine.

- Reels and time display: tape motion, timecode, varispeed, and tape speed.
- Transport: `REW`, `FF`, `STOP`, `PLAY`, and `REC`.
- Punch and marker section: `IN`, `PUNCH`, `OUT`, `REC SAFE`, `SET MK`, and `MK STOP`.
- Send selector buttons: `S1`, `S2`, `S3`, and `S4`.
- Channel strips: 16 mono tracks with meters, input labels, arm/solo/mute buttons, pan, fader, FX access, send levels, and routing.
- Bus and master area: bus returns, master level, meters, and FX.
- Side panels: tape library, audio settings, calibration/preferences, plugins, metronome, MIDI/controllers, remote, support, legal, and bounce/export.

5. Creating And Loading Tapes

Create A New Tape

- `Blank Tape`
- `Use Current Tape Settings`
- a saved template
- `16 BIT PCM`
- `24 BIT PCM`
- `32 BIT FLOAT (Recommended)`

1. Choose `New Tape`.
2. Enter a tape name.
3. Choose a starting point:
4. Choose record bit depth:
5. Press `Create`.

New tapes start at the beginning of the timeline and default the metronome to 120 BPM.

Load A Tape

Use `Load Tape` and select an existing tape session folder. TAPE 16 restores track names, levels, pan, routing, sends, markers, punch points, tape speed, and supported plugin state.

Save A Copy

Use `Save As Copy` to export the current tape as a separate session bundle. This is the safest way to make a backup before destructive recording, cleanup, imports, or major routing changes.

Templates

Use `Save As Template` to store a reusable tape setup. Templates can include routing, names, plugin assignments, mixer state, and other session settings.

Tape Library Organization

The tape library can also be used to keep sessions organized.

- Use `NEW FOLDER` to create folders inside the tape library.
- Drag a tape onto a folder to move it there.
- Use `TAPES ROOT` to choose or reveal the main library location.
- Use the sort controls to change tape order.
- Right-click or open a tape row menu for actions such as load, reveal, delete, remove missing entries, or set a progress tag.

6. Audio Setup

Open `Audio Settings` to configure recording and playback.

Important controls:

- `Audio system` (Windows): selects the driver system, such as ASIO or Windows Audio.
- `Output`: audio output device.
- `Input`: audio input device.
- `Sample rate`: current device sample rate.
- `Audio buffer size`: lower values reduce latency but increase CPU risk.
- `Test`: plays a test sound through the selected output.
- `Record Bit Depth`: current tape recording format.
- `Mix Out`: hardware output pair for the master mix.
- `Active output channels`: enabled hardware outputs.
- `Active input channels`: enabled hardware inputs.
- `Track Output Routing`: output pairs made available for per-track routing.
- `Desktop Input`: enables macOS desktop audio capture as track input choices.
- `Print Instrument To Tape As`: chooses how stereo instrument output is printed to mono tape.

For stable recording, prefer one audio interface for both input and output. Separate input/output devices can drift because they use different hardware clocks. A buffer of 128 samples or higher is safer than very small buffers when recording.

Windows Audio Interface Setup (ASIO)

For the lowest practical recording and monitoring latency on Windows, install the audio interface manufacturer's current Windows driver and select its native ASIO driver in TAPE 16.

1. Install or update the Windows driver supplied by the interface manufacturer.
2. Restart Windows after the driver installation if the installer requests it.
3. Connect and power on the interface before opening TAPE 16.

4. Open `Audio Settings`.
5. Set `Audio system` to `ASIO`.
6. Select the interface manufacturer's driver under `ASIO Driver`.
7. Open `ASIO Control Panel` and begin with a 128-sample buffer. Use 256 samples if 128 produces crackles or dropouts.

For example, a Focusrite Scarlett should normally use the Focusrite USB ASIO driver rather than appearing only as separate Windows `Speakers` and `Microphone` devices. If the ASIO device list is empty, install or repair the manufacturer's driver, reconnect the interface, restart Windows, and check again.

`Windows Audio`, `Windows Audio (Exclusive Mode)`, `Windows Audio (Low Latency Mode)`, and `DirectSound` can be useful when an ASIO driver is unavailable, but they are not substitutes for a working manufacturer ASIO driver when low-latency recording is required. Their reported input and output buffers both contribute to monitoring round-trip latency.

For near-zero-latency dry monitoring, you can also use the interface's hardware `Direct Monitor` feature. Hardware direct monitoring bypasses TAPE 16's plugin path, so PFX or MFX processing will not be heard on that direct signal.

TAPE 16 can expose large interfaces and aggregate devices with up to 128 input channels. For the most stable setup, enable only the physical inputs you are actually using in the current tape. Very large aggregate devices are best treated like studio patchbays: keep the active channel list intentional, use one clock master, and avoid changing the aggregate device while a tape is recording or playing.

7. Recording Audio

1. Open `Audio Settings` and select your input/output device.
2. Enable the physical input channels you need.
3. On each track, choose an input source.
4. Press the track `A` arm button for the track you want to record.
5. Press `REC` to latch record-ready.
6. Press `PLAY` to begin recording.
7. Press `STOP` when finished.

Arm lamps blink when ready and go solid while recording. The `REC` lamp behaves the same way: blinking means record is latched, solid means TAPE 16 is actively recording.

Because recording is destructive, only armed tracks are written. Unarmed tracks play back normally unless monitoring settings say otherwise.

8. Record Safe

`REC SAFE` temporarily disarms all currently armed tracks and remembers which tracks were armed. Press `REC SAFE` again to restore those arm states.

Use it when you want to listen back, rehearse, edit routing, or adjust effects without risking an accidental overwrite.

Default shortcut: `Q`.

9. Punch Recording

Punch recording replaces a specific section of tape.

1. Move the tape position to the punch-in point.
2. Press `IN`.
3. Move to the punch-out point.
4. Press `OUT`.
5. Press `PUNCH` to enable auto-punch.
6. Arm the track or tracks.
7. Press `REC`, then `PLAY`.

TAPE 16 records only inside the punch range. `PUNCH` requires both an in and out point before it can be enabled.

Default shortcuts:

- `[` sets punch in.
- `]` sets punch out.
- `P` toggles punch.

10. Markers

Use `SET MK` to add a marker at the current position. If the tape is already on a marker, the button changes to `DEL MK`.

Use `MK STOP` to make playback stop at markers.

Default shortcuts:

- `M` sets or deletes a marker.
- `,` toggles marker stop.

11. Tape Speed And Varispeed

TAPE 16 includes a reel-panel tape speed switch:

- `7.5 IPS`
- `15 IPS`
- `30 IPS`

Tape speed affects playback-rate character and tape tone contour. Speed changes are ramped to reduce abrupt jumps.

The `VARISPEED` control changes playback speed around the selected tape speed. If tape-speed print is enabled, speed changes can be committed into recorded audio.

12. Mixer And Tracks

Each tape track is mono by design.

Common track controls:

- `R`: arm the track for recording.
- `S`: solo.
- `M`: mute.
- Fader: track level.
- Pan: left/right placement in the master mix.
- Input label: selects the input source.
- Output/routing area: sends the track to master, bus, or hardware output routing.
- `FX`: opens plugin options for the selected track.

Double-clicking a fader resets it to `0 dB`.

Stereo Link

Tracks can be stereo-linked in adjacent pairs:

- 1-2
- 3-4
- 5-6
- 7-8
- 9-10
- 11-12
- 13-14
- 15-16

When stereo link is enabled, arm, solo, mute, and level changes can follow the pair. Tape tracks remain mono; stereo link is a workflow and routing convenience.

Track Groups

Track rename includes an optional color group. Track groups help organize related tracks and can link group volume behavior.

Bouncing Several Tracks To One Or Two Tracks

Use track bouncing when you want to commit several recorded tracks into a smaller number of tape tracks. This is useful for freeing tracks, printing a submix, or making a decision feel like part of the tape instead of leaving every layer separate.

Before bouncing, use `Save As Copy` if you may want to return to the separated tracks later. A bounce is a new recording, and recording in TAPE 16 is destructive on armed destination tracks.

To bounce several tracks down to one mono track:

1. Choose an empty destination track.
2. Set that destination track's input to `OUT`.

3. On each source track, set the output/routing area to the destination track.
4. Balance the source track faders, pan, and FX while listening.
5. Arm only the destination track.
6. Set `IN` and `OUT` if you want a fixed range.
7. Press `REC`, then `PLAY`, and let the section record onto the destination track.
8. Stop, listen back to the new track, then mute or clean out the original source tracks only when you are happy.

To bounce down to two tracks, use an adjacent pair as the destination, such as tracks 15-16, and set both destination inputs to `OUT`. Route the tracks you want on the left side to the first destination track and the tracks you want on the right side to the second destination track, then arm both destination tracks and record the pass. If you need the same source to appear on both sides of a two-track stem, print one side first, then route the source again and print the other side.

When a source track is routed directly to an armed destination track, it is removed from the normal master monitor path during that record pass and printed into the destination instead. After the record pass, TAPE 16 mutes the routed source tracks so you hear the new bounced track without doubling the original material. Unmute the source tracks if you need to revise the bounce.

13. Sends, Buses, And Master

TAPE 16 supports multiple send paths, two bus returns, master processing, and routing workflows for external/outboard gear.

- Use `S1`, `S2`, `S3`, and `S4` to select which send lane the strip controls are editing.
- Each send lane lets tracks and buses feed a shared effect path without moving the track's main output.
- Sends are useful for shared reverbs, delays, parallel compression, headphone-style cue blends, and printing several tracks through one external device.
- Send names can be renamed, so labels such as `VERB`, `DLY`, `PLATE`, or `TAPE FX` can replace generic send names.
- Bus names can be renamed for submixes such as `DRUMS`, `VOX`, or `GTRS`.
- Bus and master channels have their own level, mute/solo where applicable, meters, and FX access.

Default send shortcuts:

- `1`: select send 1.
- `2`: select send 2.
- `3`: select send 3.
- `4`: select send 4.

Using Outboard Gear

Outboard routing depends on your audio interface having enough physical outputs and inputs.

1. Open `Audio Settings`.
2. Enable the hardware outputs and return inputs you plan to use.
3. Use `Track Output Routing` and per-track output choices to send a track, bus, or send path to a physical output.
4. Patch that interface output into the input of your external hardware.
5. Patch the hardware output back into an interface input.
6. Select that return input on a spare tape track if you want to print the hardware return.

7. Arm the return track and record it back to tape.

Keep levels conservative when using external gear. If you are sending audio out and returning it to an armed track, avoid monitoring loops: do not route the return track back into the same output path that is feeding the hardware unless you intentionally want feedback.

For time-based outboard effects, print the return to tape when you are happy with it. For compression, EQ, saturation, or other insert-style processing, printing the return gives you a committed tape track that behaves like any other recorded source.

You can patch matching channel numbers when your interface workflow calls for it, such as output 1 into input 1, or a stereo pair such as outputs 1-2 returning to inputs 1-2. This is useful for hardware inserts and simple interface labeling, but it also makes feedback loops easier to create. Before arming the return track, confirm that the return is not routed back to the same physical output that is feeding the hardware.

Hardware Insert FX Slots

Some FX slots can be switched from a plugin to a hardware insert. A hardware insert sends audio out of your interface, receives it back on an input, and keeps that patch inside the tape session.

Hardware insert controls can include:

- output target,
- return input,
- latency in samples,
- wet/dry mix,
- output gain,
- input gain,
- insert name,
- enabled/bypassed state.

Use `Save Preset` for hardware patches you use often. Use `Manage Presets` to load or delete saved insert presets. Keep hardware insert levels conservative, especially when the insert return is routed back into the same session.

14. Plugins

Open `Plugin Manager` to scan and manage plugins.

Plugin Manager controls:

- `Add Folder`: add a plugin scan path.
- `Scan New`: scan normal plugin locations and added folders for plugins that are not already cached.
- `Deep Rescan`: run a full, more thorough scan when a plugin is missing or a shell/container plugin needs to be refreshed.
- `UNSCAN (Clear Cache)`: clear the plugin cache and rebuild from scratch.
- `Enable All`: re-enable skipped plugins.
- `Disable All`: skip visible plugins in the library list.
- Search field: filter plugins by name.
- Active plugin grid: edit, remove, duplicate, or replace loaded plugins.

Track FX lanes:

- `PFX`: print FX. These are printed to tape while recording.
- `MFX`: mix FX. These affect monitoring/playback and bounces without altering the recorded tape files.

Use PFX when you want to commit the effect. Use MFX when you want to keep the recorded tape dry.

TAPE 16 forwards transport, tempo, and playhead position to hosted plugins when supported. Crash-protected plugins may be kept as offline placeholders during recovery so their slots and saved state are preserved.

Moving Or Duplicating Track Plugins

In the active plugin grid, track plugins can be dragged between track FX slots.

- Drag a plugin to another track or slot to move it.
- Hold `Control` while dropping to duplicate it instead.
- A duplicated plugin keeps the same plugin state and bypass state when the plugin reports its state correctly.
- Moving a plugin removes it from the original slot after the destination loads.

This drag workflow is for track plugin slots. Send, bus, and master slots can still be changed from their own plugin slot controls.

Plugin Library Filters

The plugin library can hide whole formats or individual plugins.

- `No AU` hides Audio Unit plugins.
- `No VST3` hides VST3 plugins.
- `All On` clears skipped plugins and formats.
- `All Off` skips the visible plugin list.

If a plugin disappears, check the search field, format filters, and skipped plugin state before doing a full rescan.

Red active slots indicate high-latency plugins. They can still be useful while mixing, but they are less suitable for low-buffer tracking, reel slowdown, or heavy wow/flutter use.

Plugin State And Tape Sessions

Plugin assignments and supported plugin state are saved with the tape session. TAPE 16 writes a `plugin_state.xml` file plus a `plugin_state` sidecar folder inside the tape folder so larger plugin states can be stored outside the main tape state file. This is the format used for complex instruments and samplers when they report their state correctly.

State is saved during normal session handoff, including loading another tape, saving a copy, saving a template, and closing or switching sessions. For the safest recall with large instruments such as drum instruments, samplers, or multi-output instruments, let the plugin finish loading its kit/library before switching tapes, and avoid quitting while the plugin is still downloading, scanning, or showing a modal login window.

If a plugin cannot be loaded during crash recovery or safe-mode loading, TAPE 16 may keep the slot as an offline placeholder so the assignment and saved state are not immediately lost. Re-scan or re-enable the plugin, then reload the tape to restore it.

Use `Restore Backup` in Plugin Manager if a recent plugin state save looks wrong or a tape opens with missing plugin state after a crash. Use `Reset Plugin System` only when scanning or plugin loading is badly stuck, because it is a broader recovery action.

VST3 Scanning And Waves-Style Shells

On macOS, TAPE 16 scans the standard VST3 locations:

- `/Library/Audio/Plug-Ins/VST3`
- `~/Library/Audio/Plug-Ins/VST3`

On Windows, the standard 64-bit VST3 location is:

- `C:\Program Files\Common Files\VST3`

Make sure the VST3 edition of each plugin is installed. Installing only an AAX or other plugin-format edition will not place a VST3 in the standard folder. If a manufacturer installs its VST3 files in a custom location, use `Add Folder` to include that location.

Use `Add Folder` if a plugin installer placed VST3 files somewhere else. Use `Scan New` after installing normal one-plugin-per-file plugins. Use `Deep Rescan` when a plugin does not appear after scanning new plugins. Deep Rescan is slower, but it can discover plugins that are exposed through a container or shell rather than a simple one-plugin-per-file layout.

Some manufacturers, including Waves, can expose plugins through a WaveShell-style VST3 container. In that case the scanner may need to probe the shell before the individual plugin names appear in the library. If Waves plugins still do not appear, confirm that the VST3 versions are installed in Waves Central, clear the TAPE 16 plugin cache with `UNSCAN (Clear Cache)`, then run `Deep Rescan`.

Typing Into Plugin Windows

Plugin editors can contain login forms, search fields, serial-number fields, and preset browsers. TAPE 16 keeps normal printable typing inside the plugin window so those fields can receive text. Transport shortcuts still work from the main TAPE 16 window, and command-style shortcuts such as closing a plugin window remain available where supported.

15. MIDI And Instrument Recording

TAPE 16 supports MIDI input and instrument plugin workflows.

Track input choices can include:

- audio inputs,
- desktop audio left/right,
- MIDI,
- MIDI channel choices,
- stereo MIDI print modes using an adjacent track for the right side.

In `Audio Settings`, `Print Instrument To Tape As` controls how a stereo instrument is printed to a mono tape track:

- `MONO SUM (L+R)`
- `LEFT ONLY`
- `RIGHT ONLY`

Use `MIDI Panic` if held notes or a plugin instrument get stuck.

16. MIDI Clock And MTC

Open the MIDI/controller settings to configure sync outputs:

- aggregate MIDI inputs,
- aggregate MIDI outputs,
- MIDI clock output,
- MTC or MIDI Clock + MTC sync output.

The sync master path can send from the virtual `TAPE 16 Sync` MIDI output. Use `MTC Only` when a DAW such as Logic Pro should chase TAPE 16's timeline. Use `MIDI Clock + MTC` when you also need drum machines, arpeggiators, or other hardware to receive tempo clock.

The MIDI clock option sends 24 PPQN MIDI Clock from the metronome BPM. Enable `Send MIDI Clock` and select the MIDI output device that should receive clock.

Use `Clock Offset` in the `SYNC CLOCK` tab to fine-tune external drum machines and sequencers. Positive values send MIDI Clock earlier, while negative values delay it.

Sync Outputs

The `SYNC CLOCK` tab lets TAPE 16 act as a sync master and send MIDI Clock, MTC, or both to selected MIDI outputs. `Sync Outputs` is a per-port list, so you can send sync to only the drum machine, interface MIDI port, IAC bus, or DAW destination that needs it.

When troubleshooting sync, enable one output first. Duplicate MIDI Clock or MTC on several ports can make external devices appear late, jittery, or double-triggered if the same destination receives the signal twice.

17. DAW Controller Setup

The controller panel includes Mackie Control/MCU-oriented configuration, SSL UF8 send/plugin button behavior, and MIDI Learn maps for assigning hardware controls.

Supported controller profiles include Mackie Control, PreSonus FaderPort, Behringer X-Touch, Softube Console 1, and HUI-compatible devices. Add multi-unit Mackie-style controllers in the same left-to-right order as the physical hardware.

The Mackie Control and Mackie-compatible setup is the main supported controller path. Behringer X-Touch units should normally use the Mackie-compatible profile. Softube Console 1, HUI-compatible controllers, and PreSonus FaderPort support are available for testing, but may need extra verification with your exact hardware and firmware.

Controller Setup

Connect and power on the controller before opening TAPE 16. If the controller has a DAW mode, set it to Mackie Control, MCU, Logic Control, or a Mackie-compatible mode before assigning it in TAPE 16.

In the MIDI/controller settings, open the `DAW Controllers` tab and add a controller entry for each physical Mackie-compatible MIDI port you want TAPE 16 to use. Choose the matching profile, then select that unit's MIDI input and MIDI output. The input receives fader, knob, and button moves from the hardware; the output sends LEDs, meters, scribble strips, time display, and fader feedback back to the hardware.

For a single Mackie-compatible 8-fader surface, add one controller entry and select its matching input/output pair. For a main unit plus extenders, add the main unit first, then add each extender in the same left-to-right order as the physical hardware. Keep each unit's input and output paired to the same physical controller.

After assigning ports, move a fader, press select, or bank the controller to confirm TAPE 16 is receiving input. Then arm a track or press transport buttons to confirm LED feedback is returning to the surface. If the controller does not respond, check that no other DAW is already using the same MIDI ports, then disable and re-enable the controller entry.

Mackie Control Operation

On Mackie Control compatible surfaces, the channel `REC/RDY` or record-arm buttons follow the visible TAPE 16 record-arm state. Armed tracks flash at the same pace as the on-screen TAPE 16 record-arm lights, then stay solid while TAPE 16 is actively recording.

The Mackie transport record button follows the same behavior as TAPE 16's record button. It flashes while record is latched and stays solid during active recording.

In the normal Mackie pan mode, turning a channel VPot adjusts that track's pan. Pressing the VPot resets the mapped track pan to center. This reset applies to the track currently assigned to that surface channel, including the current bank or selected-track mapping.

Mackie Control Extenders

Mackie Control compatible controllers can be grouped into a wider surface. In the `DAW Controllers` tab, add the main Mackie Control unit first, then add extenders in the physical left-to-right order you want to use.

For Mackie-style 8-fader units:

- first enabled unit: strips 1-8,
- second enabled unit: strips 9-16,
- additional enabled units continue the same pattern.

Bank moves stay inside the 16 TAPE tracks until the final track bank. One 8-fader unit banks between tracks 1-8 and 9-16. Two 8-fader units already cover tracks 1-16. Press `Bank Right` from the final track bank to open the `OUTPUTS` page, where the first three banked faders control `BUS 1`, `BUS 2`, and `MSTR`; remaining faders are blank. Press `Bank Left` to return to the final track bank. Channel moves still step by one track until the edge of the 16-track layout and do not enter the `OUTPUTS` page. If a controller has 10, 12, 16, or 24 physical faders but presents itself as multiple Mackie-compatible ports, add each port in the order that matches the hardware layout.

MIDI Learn

MIDI Learn lets you map knobs, faders, buttons, notes, pitch wheel, or controller messages from external MIDI hardware to TAPE 16 controls.

Common MIDI Learn targets include:

- track level, pan, arm, solo, and mute,
- bus level and master level,
- varispeed,
- transport controls,
- `REC SAFE`,
- punch in, punch enable, and punch out,
- send selection,
- marker set and marker stop,
- tape-character controls such as wow, flutter, saturation, tape age, tape damage, and tape echo.

Basic workflow:

1. Press `L` or enable `MIDI LEARN`.

2. Click or select the TAPE 16 control you want to map.
3. Move the hardware control you want assigned.
4. Test the control and adjust your hardware mode if the response is inverted, too sensitive, or momentary when you expected toggle behavior.

Use `CLEAR` beside an individual mapping to remove it, or `CLEAR ALL` in the MIDI Learn map window to start over. If a controller sends the same MIDI CC on several knobs or faders, change the controller's MIDI template first so each hardware control sends a unique message.

Use `SAVE PROFILE` to back up or share MIDI Learn mappings. Use `LOAD PROFILE` to replace the current map with a saved profile. Use `MERGE` to add mappings from a profile while keeping your current map; if there are conflicts, TAPE 16 asks how to handle them.

Default shortcut for MIDI Learn: `L`.

18. Metronome

Open the metronome window to control:

- on/off,
- BPM,
- tap tempo,
- time signature,
- `Play only on record`,
- `Accent first beat`,
- semitone,
- level.

Default shortcut: `C`.

The metronome BPM is also used for MIDI Clock output.

19. Remote Control

TAPE 16 includes a local phone/tablet remote.

The computer running TAPE 16 and the phone/tablet remote must be on the same Wi-Fi network. The remote is local to your network; it is not a cloud control page.

1. Open the remote panel in TAPE 16.
2. Scan the QR code with a phone on the same Wi-Fi network.
3. Use the browser remote to arm tracks, control transport, set markers, toggle record safe, toggle click, and control punch.

The remote shows:

- connection status,
- track arm buttons 1-16,
- timecode,
- marker state,

- punch in/out readouts,
- transport buttons.

For the cleanest phone experience, add the remote page to your home screen.

20. Importing WAV Files

Use `Import` to place WAV audio onto the tape.

Options:

- choose the destination track,
- enable `Import as stereo pair`,
- set the start time in seconds,
- press `Import`.

Imported audio is written into the selected tape track range, so treat it like a tape operation.

21. Erasing And Clean Out

Use `Erase` to erase a selected time range from selected tracks.

Options:

- select tracks individually,
- `ALL`,
- `NONE`,
- `Erase Range`,
- `Clean Out`.

`Erase Range` inserts blank audio only between `IN` and `OUT` on checked tracks. `Clean Out` removes each checked track and moves its WAV to Trash. Make a `Save As Copy` first if you may need to recover the material.

22. Bounce And Export

TAPE 16 writes bounce files as `.wav`.

Range options:

- punch range: requires both `IN` and `OUT`,
- full tape: uses the recorded tape length.

Format options:

- `32-bit float`,
- `24-bit PCM`,
- `16-bit PCM`.

Sample-rate options may include:

- current session/device rate,

- `44.1 kHz`,
- `48 kHz`,
- `96 kHz`.

Realtime bounce records exactly what you hear and stops automatically at the selected out/end point when using a fixed range. Offline bounce may be hidden or unavailable in some builds.

Use `OPEN FOLDER` in the bounce panel to open the current tape's audio files folder. After a realtime bounce finishes, TAPE 16 also reveals the saved bounce location when possible.

23. Keyboard Shortcuts

Open `Shortcuts` to edit key assignments. Each action has two shortcut slots.

Default shortcuts:

Action	Default
Rewind	Left Arrow
Fast Forward	Right Arrow
Stop	Down Arrow
Play	Up Arrow
Reverse	Z
Play / Stop	Space
Record	R
Rec Safe	Q
Set Marker	M
Marker Stop	,
Punch In	[
Punch	P
Punch Out	]
Send S1	1
Send S2	2
Send S3	3
Send S4	4
Metronome	C

MIDI Learn	L
Plugins	Backtick

Hold `Shift` while rewinding or fast-forwarding to use 10x shuttle speed.

W, A, S, and D are not assigned to transport by default so plugin login windows, preset search fields, and text-entry controls can receive normal typing. If you assign letter keys manually, remember that those keys may be captured when focus is in the main TAPE 16 window.

24. Preferences And Calibration

The calibration/preferences area controls visual and workflow preferences such as:

- UI size,
- text size,
- VU height,
- reel size,
- side menu button size,
- UI FPS: 24, 30, or 60,
- fullscreen behavior,
- tape library sort mode,
- meter glow,
- shadows and visual offsets,
- input monitoring,
- playback while record-armed.

Preferences are stored in your local `TapeDAW` app data folder.

Useful preference notes:

continues to control whether the live input is heard alongside it; recording replaces armed-track tape with live input.

- `OPEN TAPE 16 WITH` can start with a new tape, the last tape, or a selected template.
- `PLUGIN EDITOR DISPLAY` chooses where plugin windows should open on multi-monitor systems.
- `INPUT MONITOR` controls live input monitoring.
- `PLAY WHILE ARMED` keeps existing tape audible on armed tracks while the transport is playing. `INPUT MONITOR`
- `CHECK UPDATE` checks the public release page.
- `AUTO UPDATE` allows TAPE 16 to periodically check for new releases.

25. Themes

TAPE 16 themes let you customize the machine's visual style without replacing the bundled app files. A theme can store replacement PNG textures, color changes, font choices, and small positioning or tint adjustments for supported artwork slots.

Open the calibration/preferences side panel and press `THEMES`. The dialog is called `TAPE 16 Themes`.

Creating And Applying A Theme

1. Open the theme manager.
2. Choose `NEW` to create a local theme.
3. Enter a theme name, author, and version.
4. Choose a slot from the texture, color, or font list.
5. Use `REPLACE IMAGE`, adjust the tint/brightness/scale/offset, or choose the color/font setting you want.
6. Use `APPLY` to make the selected theme active.

Theme edits auto-save as you work. Use `RESET SLOT` to return one texture, color, or font slot to its default state. Use `RESET ACTIVE` to clear the active theme and return the app to the built-in look.

Use `Hide selected element` on a texture slot when you want that UI element to disappear in the active theme. For example, select `TAPE 16 Logo` and turn on `Hide selected element` to remove the logo while keeping the rest of the panel visible.

Replacing Texture PNGs

Texture slots accept PNG files. For the cleanest result, make replacement images the same pixel size as the bundled source image for that slot. TAPE 16 can scale and offset texture slots, but matching the original size avoids unexpected stretching, cropping, or softness.

Some slots are tiled textures, such as wood, reel tape, button texture, and metal panels. These usually work best when the replacement PNG is seamless. Other slots are fixed sprites, such as R/S/M buttons, speed-switch positions, meter segments, and meter masks. Fixed sprites should match the source dimensions exactly.

The theme editor also includes brightness, opacity, tint, scale, and X/Y offset controls. These are useful for small adjustments after importing a PNG. If a replacement already has the correct brightness, color, and visibility, leave opacity and scale at `1.00` and tint amount at zero.

Reel PNG Alpha

When a reel metal face slot is selected, the `Use reel PNG alpha` checkbox appears with the texture controls. It is off by default so older themes keep TAPE 16's built-in reel hole shapes.

Replace `Front Reel Metal Face` or `Back Reel Metal Face`, then turn `Use reel PNG alpha` on when you want that reel PNG to define the reel face. TAPE 16 still clips the outside edge to a circle, but it stops cutting the built-in reel holes. Transparent areas in the reel PNG show the tape behind the reels; solid areas stay solid, so a full solid reel image has no window holes.

Enable `Advanced PNG slots` if you want separate supply and takeup reel artwork. The advanced reel overrides are `Left Back Reel Metal Face`, `Right Back Reel Metal Face`, `Left Front Reel Metal Face`, and `Right Front Reel Metal Face`. If a left/right slot is empty, TAPE 16 falls back to the matching shared front or back reel face.

Editing Colors And Fonts

Color slots control text, meter, LED, and machine accent colors. Select a color slot, enter a hex color, or use `PICK` to choose a color visually.

Font slots control groups of machine text rather than every label individually. Built-in choices include segmented, system, and condensed styles. Choose `custom` and import a `.ttf` or `.otf` file if you want a theme to carry its own font.

Advanced PNG Slots

By default, the slot list shows the main curated texture slots. Enable `Advanced PNG slots` to reveal smaller support sprites such as button halos, meter glow sprites, and meter masks.

Advanced slots are powerful but less forgiving. Treat them as exact artwork replacements: keep their dimensions, transparency, and alignment as close to the original as possible.

Importing And Exporting Themes

Theme packages can be exported as ZIP files and imported again later. This is the safest way to back up a custom theme, move it between machines, or share it with another TAPE 16 user.

When sharing a theme, include a clear name, author, and version so future edits are easy to identify. If you are experimenting heavily, export a backup before replacing many texture slots.

Themes are stored in the local `TapeDAW/Themes` app data folder. Use `REVEAL FOLDER` in the theme manager if you need to inspect or back up the theme files directly.

Default Texture Slot Sizes

These texture slots are visible in the normal theme slot list.

Theme slot	Recommended PNG size
TAPE 16 Logo	1375 x 432
Black Plastic Panel	1536 x 1024
Logo Plate	1536 x 1024
Back Reel Metal Face	1536 x 1024
Front Reel Metal Face	1536 x 1024
Reel Hub Brushed Metal	1536 x 1024
Reel Pin / Hub Cap	512 x 512 transparent PNG
Reel Tape	1024 x 1024
Wood Faceplate	1024 x 1024
Fader Bay Metal Texture	1536 x 1024
Round Knob/Button Texture	1024 x 1024
Record/Solo/Mute Button Off	36 x 22

Record Button On	36 x 22
Solo Button On	36 x 22
Mute Button On	36 x 22
Meter Segment Off	28 x 10
Meter Segment Green	28 x 10
Meter Segment Yellow	28 x 10
Meter Segment Red	28 x 10
Speed Switch 7.5 IPS	462 x 190
Speed Switch 15 IPS	471 x 192
Speed Switch 30 IPS	462 x 189

Advanced Texture Slot Sizes

These slots appear when `Advanced PNG slots` is enabled.

Theme slot	Recommended PNG size
Record Button Halo	64 x 39
Solo Button Halo	64 x 39
Mute Button Halo	64 x 39
Meter Segment Glow Green	60 x 30
Meter Segment Glow Yellow	60 x 30
Meter Segment Glow Red	60 x 30
Meter Mask: Glow Containment	96 x 360
Meter Mask: Strong Glow Containment	96 x 360
Meter Mask: Inner Clip	96 x 360

26. Support And Crash Recovery

Use `Report Bug / Crash` to send a report from inside the app. Include:

- what happened,
- exact steps,
- audio device,

- sample rate,
- buffer size,
- plugins involved.

The report includes audio diagnostics and a support bundle when available. Startup crash reports may include machine/runtime details and attachment hints for diagnostics.

If TAPE 16 detects an unclean shutdown, it can open a crash recovery flow and create or load a safe-mode copy of a tape. Safe-mode tapes are named with a `Safe Mode` suffix. Exit recovery mode when you are ready to load tapes normally again.

27. Practical Workflows

Record A Vocal

1. Create or load a tape.
2. Open `Audio Settings` and select your interface.
3. Enable the microphone input channel.
4. Pick that input on a track.
5. Arm the track.
6. Press `REC`, then `PLAY`.
7. Press `STOP` to finish.

Commit An Effect To Tape

1. Load the plugin in the track `PFX` lane.
2. Arm the track.
3. Record through it.

Keep An Effect Flexible

1. Load the plugin in the track `MFX`, bus, send, or master lane.
2. Leave the recorded tape dry.
3. Adjust or remove the effect later.

Print Outboard Gear

1. Route a track, send, or bus to a hardware output.
2. Patch that output through the external processor.
3. Return the processor to an interface input.
4. Select the return input on a spare tape track.
5. Arm and record the return.

Export A Mix

1. Set `IN` and `OUT`, or choose full tape.
2. Choose the bounce format.
3. Start realtime bounce.
4. Let playback run until the selected end point.

Bounce Tracks Down

1. Save a copy of the tape if you want a backup of the separated tracks.
2. Choose one spare track, or an adjacent pair for a two-track bounce.
3. Set the destination track input to `OUT`, or set both destination inputs to `OUT` for a two-track bounce.
4. Route the source tracks to the destination track or tracks.
5. Arm only the destination track or pair.
6. Record the pass, then listen back before muting or cleaning out the source tracks.

28. Sync Logic Pro To TAPE 16

Use this when TAPE 16 should be the master transport and Logic Pro should chase TAPE 16's timecode.

TAPE 16 Settings

1. Open `MIDI I/O`.
2. Open the `SYNC CLOCK` tab.
3. Turn on `Enable Sync Master`.
4. Set `Format` to `MTC Only`.
5. Set `MTC Rate` to match the Logic project frame rate.
6. In `Sync Outputs`, enable only the virtual `TAPE 16 Sync` output while testing.

Use `MIDI Clock + MTC` only if you also need external drum machines, arpeggiators, or MIDI hardware to receive MIDI Clock from TAPE 16.

Logic Pro Settings

- `Bar Position: 1 1 1 1`
- `plays at SMPTE: 00:00:00:00`

1. Open `File > Project Settings > Synchronization`.
2. In the `General` tab, set Logic to chase external `MTC`.
3. Set Logic's frame rate to the same rate selected in TAPE 16.
4. Set Logic's SMPTE offset so the project starts at zero:
5. Enable Logic's sync/external sync button in the transport.
6. Press play in TAPE 16. Logic should chase TAPE 16's timecode.

Logic commonly defaults bar 1 to `01:00:00:00`. If that is left unchanged, incoming TAPE 16 timecode such as `00:00:32:00` appears before bar 1 in Logic, which can make the playhead look like it is behind the visible timeline.

Troubleshooting Logic Sync

If Logic plays at the same time but the timeline is far behind bar 1:

- Recheck `Bar Position: 1 1 1 1 plays at SMPTE: 00:00:00:00`.
- Recheck that both apps use the same MTC frame rate.

If Logic moves in time but jitters or pulls back and forth:

- Use `MTC Only` in TAPE 16 while testing.
- Enable only one TAPE 16 sync output, preferably the virtual `TAPE 16 Sync` output.
- Make sure Logic is not also sending MIDI Clock or MTC back to TAPE 16.
- Make sure Logic is not receiving MTC from another device, IAC bus, controller, or interface MIDI port.
- Disable automatic MTC frame-rate detection in Logic if it keeps hunting, then set the frame rate manually to match TAPE 16.

For reliable timecode lock, TAPE 16 and Logic should use the same audio sample rate. MTC synchronizes transport position/timecode; it does not replace a shared audio clock.

29. Sync Ableton Live To TAPE 16

Use this when TAPE 16 should be the master transport and Ableton Live should follow TAPE 16's timecode.

TAPE 16 Settings

1. Open `MIDI I/O`.
2. Open the `SYNC CLOCK` tab.
3. Turn on `Enable Sync Master`.
4. Set `Format` to `MTC Only`.
5. Set `MTC Rate` to match the frame rate you will choose in Ableton Live. `30 fps` is a good default if the Ableton set does not already require another rate.
6. In `Sync Outputs`, enable only the virtual `TAPE 16 Sync` output while testing.

Use `MIDI Clock + MTC` only if you also need external drum machines, arpeggiators, or MIDI hardware to receive MIDI Clock from TAPE 16.

Ableton Live Settings

1. Open `Live > Settings` or `Live > Preferences`.
2. Open the `Link, Tempo & MIDI` tab.
3. In `Input Ports`, find `TAPE 16 Sync`.
4. Turn on `Sync` for the `TAPE 16 Sync` input. Leave `Track` and `Remote` off unless you also need MIDI notes, clips, or controller messages from that port.
5. Expand the `TAPE 16 Sync` input port settings.
6. Set `Sync Type` to `MIDI Timecode`.

7. Set `MTC Frame Rate` to the same rate selected in TAPE 16. If you used the recommended default, choose `30 fps`.
8. Set `MTC Start Offset` to `00:00:00:00`.
9. Enable Ableton Live's `Ext` button in the top transport bar.
10. Press play in TAPE 16. Ableton Live should chase TAPE 16's timecode.

Troubleshooting Ableton Sync

If Ableton Live does not move when TAPE 16 plays:

- Check that Ableton Live's `Ext` button is enabled.
- Check that `Sync` is enabled for the `TAPE 16 Sync` input port.
- Check that `Sync Type` is set to `MIDI Timecode`, not `MIDI Clock`.
- Check that the same MTC frame rate is selected in both apps.

If Ableton Live starts from the wrong position:

- Set `MTC Start Offset` to `00:00:00:00`.
- Make sure TAPE 16 is sending MTC from only one output while testing.
- Disable other MIDI sync sources in Ableton Live that might also be sending clock or timecode.

For reliable timecode lock, TAPE 16 and Ableton Live should use the same audio sample rate. MTC synchronizes transport position/timecode; it does not replace a shared audio clock.

30. Troubleshooting

No Input Signal

- Check `Audio Settings`.
- Make sure the input device is selected.
- Enable the physical input channel.
- Check the track input label.
- Arm the track.
- Confirm macOS microphone permission if prompted.

Crackles Or High-Pitched Noise

- Increase buffer size to 128 samples or higher.
- Use one interface for both input and output.
- Avoid recording with separate input/output devices unless you know they are clocked together.
- Reduce plugin load while tracking.
- At 88.2 kHz or 96 kHz, use a larger buffer when possible, especially with heavy plugins or instruments.
- If 96 kHz feels unstable, test the same tape at 44.1 kHz or 48 kHz to separate sample-rate load from routing or plugin problems.

High Monitoring Latency On Windows

- Open `Audio Settings` and select `ASIO` as the audio system.
- Select the native ASIO driver supplied by the audio interface manufacturer.
- If no ASIO driver is listed, install or repair the manufacturer's current Windows driver, reconnect the interface, and restart Windows.
- Open `ASIO Control Panel` and try 128 samples. Use 256 samples if the lower setting is not stable.
- Avoid generic Windows `Speakers` and `Microphone` device pairs when the interface has a native ASIO driver.
- Bypass high-latency plugins while tracking. Red plugin slots indicate plugins that report high latency.
- Use the interface's hardware direct-monitoring feature when you need the lowest possible dry-monitoring latency.

Plugin Not Found

- Open `Plugin Manager`.
- Add the plugin folder if needed.
- Run `Scan New`.
- Use `Deep Rescan` if the normal scan misses it.
- Check that the plugin has not been disabled or skipped.
- For Waves or other shell/container plugins, make sure the VST3 version is installed, clear the scan cache with `UNSCAN (Clear Cache)`, then run `Deep Rescan`.
- On Windows, check the standard 64-bit VST3 location: `C:\Program Files\Common Files\VST3`.
- Check the standard macOS VST3 locations: `/Library/Audio/Plug-Ins/VST3` and `~/Library/Audio/Plug-Ins/VST3`.

Stuck MIDI Notes

- Use `MIDI Panic`.
- Check MIDI input routing.
- Reload the instrument plugin if needed.

Remote Does Not Connect

- Make sure the phone and computer are on the same Wi-Fi network.
- Keep TAPE 16 open.
- Rescan the QR code.
- Check that local network/firewall settings are not blocking the remote server.

Bounce Is Unavailable

- Make sure the tape has recorded audio.
- If using punch range, set both `IN` and `OUT`.

31. Safety Notes

- Recording is destructive on armed tracks.
- Import and erase operations alter tape contents.
- PFX plugins are printed to tape while recording.
- MFX, bus, send, and master FX remain mix-time processing.
- Outboard returns can feed back if routed incorrectly.
- Use `Save As Copy` before destructive edits, punch-ins, cleanup, or experimental routing.